

L -functions for all meromorphic $\mathrm{SL}_2(\mathbb{Z})$ modular forms, via manifestly finite contours

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Abstract

We give a single manifestly finite, regulator-free functional realizing in one object L -functions quasi-periods and period polynomials of meromorphic modular forms on $\mathrm{SL}_2(\mathbb{Z})$, with generic poles throughout the upper half-plane $\mathbb{H}$ and at the cusp of $\mathrm{SL}_2(\mathbb{Z})$. Additionally, our construction preserves holomorphicity. It is a two-segment contour integral, $L^*(f, s) = L^*(f, s; \gamma_{\tau_0}^T) + L^*(f, s; \gamma_{\tau_0}^S)$, where the $\gamma_{\tau_0}^S$ is a path in $\mathbb{H}$ from $S(\tau_0)$ to τ_0 , and the same for $\gamma_{\tau_0}^T$. For weakly holomorphic forms our L -function is independent of τ_0 and of the path, and reproduces classical L -functions for holomorphic modular forms, reproduces the L -functions of weak modular forms in [1], which previously needed non-holomorphic regularization to be defined, and reproduces Brown's quasi-periods [2]. For properly meromorphic modular forms with poles inside the interior of $\mathbb{H}$, our L -function depends on the homotopy classes, defined in terms of pole-crossings. There is a natural choice of trivial homotopy class, which renders our L -functions canonical when the modular form has a pole in the interior of the fundamental domain, generalizing [3].

1 Introduction and preliminary concepts

Our primary goal in this note is to put forth a unifying generalization of the L -function associated to a given modular form. Our L -function naturally encompasses the classical case of L -functions associated to holomorphic modular forms [4], and seamlessly unifies Ref. [1]'s extension to L -functions of modular forms with poles at the cusp with Ref. [3]'s extension to L -functions of modular forms with poles at finite locations, exclusively away from cusps. Secondly, we show that our L -functions reproduce the numerical results for quasi-periods associated to weak cusp forms [2].

Note: Throughout this document, we refer to L -integrals and L -functions. The classical L -function-as-Dirichlet-series which has been suitably extended to the complex s plane is simply not defined when the L -function in question has poles either at the cusp or at finite locations within the upper half-plane. So, if there is ever a doubt, we state clearly: *any L -function in this note is defined by an L -integral*. The focus of this note is to define L -integrals for any integer weight $\mathrm{SL}_2(\mathbb{Z})$ modular form which may have poles at cusps or away from cusps.

1.1 Modular forms

Write $\mathrm{Re}(z)$ and $\mathrm{Im}(z)$ for the real and imaginary parts of a complex number $z \in \mathbb{C}$. Let $\mathbb{H} := \{\tau \in \mathbb{C} \mid \mathrm{Im}(\tau) > 0\}$ denote the upper half-plane. $\mathrm{SL}_2(\mathbb{Z})$ is the group generated by $T := \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ and $S := \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$.

A *holomorphic modular form* of weight k for $\mathrm{SL}_2(\mathbb{Z})$ is a holomorphic function $f : \mathbb{H} \rightarrow \mathbb{C}$ such that $f(\tau+1) = f(-\frac{1}{\tau})\tau^{-k} = f(\tau)$ for all $\tau \in \mathbb{H}$, and $f(\tau) = \mathcal{O}(1)$ as $\mathrm{Im}(\tau) \rightarrow \infty$, and which has no poles at finite locations within $\mathbb{H}$. By virtue of the periodicity $f(\tau) = f(\tau+1)$ and holomorphicity,

each holomorphic modular form has a q-series decomposition,

$$f(\tau) = \sum_n c_f(n) e^{2\pi i n \tau} = \sum_n c_f(n) q^n \quad , \quad q := e^{2\pi i \tau} \quad . \quad (1)$$

In order for f to be bounded as $\tau \rightarrow i\infty$, we conclude that $c_f(n) = 0$ for all $n < 0$, i.e. that $f = \sum_{n \geq 0} c_f(n) q^n$; for such modular forms f we write $f \in M_k$. If a modular form f additionally has a vanishing constant term, i.e. if $c_f(0) = 0$, then f is a *cusp form*, and we write $f \in S_k$. Clearly, $S_k \subset M_k$. Further, M_k is a finite-dimensional vector-space, as is S_k . Holomorphic $\mathrm{SL}_2(\mathbb{Z})$ modular forms must have even integer weights, and $k \geq 4$ [4].

A product of modular forms of weights k_1 and k_2 is a modular form of weight $k_1 + k_2$, so $M_* := \bigoplus_k M_k$ is naturally a (graded) ring. This graded ring is freely generated by the two modular forms $E_4 \in M_4$ and $E_6 \in M_6$, the $k = 4, 6$ specializations of the *weight k Eisenstein series* E_k , which may be defined as follows:

$$E_k := \frac{1}{2} \sum_{(m,n)=1} \frac{1}{(m+n\tau)^k} = 1 - \frac{2k}{B_k} \sum_{n=1}^{\infty} \sigma_{k-1}(n) q^n \quad , \quad (2)$$

where $\sum_{(m,n)=1}$ denotes summing over all relatively prime $(m, n) \in \mathbb{Z}^2$, the Bernoulli number B_n is the coefficient of $t^n/n!$ of the Laurent expansion of $t/(e^t - 1)$ about $t = 0$, and $\sigma_m(n) := \sum_{d|n} d^m$ is the Divisor function where $d|n$ means $n/d \in \mathbb{N}$. In short, $M_* = \mathbb{C}[E_4, E_6]$. We now define the *modular discriminant*:

$$\Delta := \frac{E_4^3 - E_6^2}{1728} = q + O(q^2) \in S_{12} \quad . \quad (3)$$

Holomorphicity is most canonically relaxed via allowing a modular form to have a pole at the cusp of $\mathrm{SL}_2(\mathbb{Z})$, i.e. for a weight k modular form for $\mathrm{SL}_2(\mathbb{Z})$ to have negative powers of q in its q-series expansion. In this case, f diverges as $\tau \rightarrow i\infty$ as $f \sim e^{-2\pi N \tau}$, where $-N$ is the largest negative number such that $c_f(n)$ is non-zero. Such a modular form f is called a *weak modular form*, and we write $f \in M_k^!$. If a weak modular form f has a vanishing coefficient of q^0 , i.e. if $c_f(0) = 0$, it is called a *weak cusp form* and we write $f \in S_k^!$. Clearly, $S_k^! \subset M_k^!$, $S_k \subset S_k^!$ and $M_k \subset M_k^!$. Two important weak cusp forms in our paper are as follows:

$$J := \frac{E_4^3}{\Delta} - 744 = \frac{1}{q} + O(q) \in S_0^! \quad , \quad (4)$$

$$\widehat{\Delta} = (J^2 + a_1 J + a_0) \Delta := \frac{1}{q} + O(q^2) \in S_{12}^! \quad , \quad (5)$$

where a_0, a_1 are integers which exist to ensure the q-series gap that defines $\widehat{\Delta}$ [5].

In this work, we further relax from even weakly holomorphic modular forms, and consider *meromorphic modular forms* of weight k which are quotients: $f = \frac{f_1}{f_2}$ where $f_i \in M_{k_i}$, the denominator f_2 is not identically zero, and $k = k_1 - k_2$. We write F_k for the vector space of meromorphic modular forms of weight k , and set $F_* := \bigoplus_k F_k$. Then F_k vanishes unless k is an even integer, and F_* may be regarded as a “graded ring of fractions” of M_* .

1.2 L-functions for holomorphic modular forms and periods for cusp forms

For any $f \in S_k$, one may define the associated L-function:

$$L(f, s) := \frac{(2\pi)^s}{\Gamma(s)} L^*(f, s) \stackrel{f \in S_k}{:=} \frac{(2\pi)^s}{\Gamma(s)} \int_0^\infty f(it) t^{s-1} dt = \sum_{n \geq 1} \frac{c_f(n)}{n^s} \quad . \quad (6)$$

A similar definition applies for $f \in M_k$. (The classical definition of $L^*(f, s)$ for $f \in M_k$ is the focus of Remark 3.6, below.) Similarly, one may define a period polynomial for $f \in S_k$ in terms of this L-function,

$$r_f(X, Y) := (2\pi i)^{n+1} \sum_{\ell=0}^n (-1)^\ell \binom{n}{\ell} i^{\ell+1} L^*(f, \ell+1) X^{n-\ell} Y^\ell, \quad n := k-2. \quad (7)$$

This polynomial, $r_f(X, Y) \in V_{k-2} := \mathbb{C}[X, Y]_{k-2}$, has a natural action under $\mathrm{SL}_2(\mathbb{Z})$, $(P|_\gamma)(X, Y) = P(aX + bY, cX + dY)$ for $\gamma = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathrm{SL}_2(\mathbb{Z})$. This period polynomial can be decomposed into a basis of highly constrained polynomials, which are annihilated by $|(1+T)$ and $|(1+U+U^2)[6]$, where $U := TS$ generates an order-3 cyclic subgroup of $\mathrm{SL}_2(\mathbb{Z})$. We define the *periods* of a cusp form to be the coefficients that localize r_f within this highly constrained sub-space of V_{k-2} .

However, as classically computed, these L-functions and associated period polynomials (when defined in terms of the special values of the associated L-function) are not defined when f has poles either at the cusp of $\mathrm{SL}_2(\mathbb{Z})$ or within the interior of $\mathbb{H}$. As stated above, there are extant, but partial and non-overlapping, extensions of the definition of $L(f, s)$ to $f \in M_k^!$ or $f \in F_k$. In this note, we provide a construction that unifies these disparate extensions.

2 A finite L-integral

Definition 2.1 (*S- and T-segments*). Let $\tau_0 \in \mathbb{H}$ and $f \in F_k$, let $S = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$, $T = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ denote the generators of $\Gamma = \mathrm{SL}_2(\mathbb{Z})$, and let $f^{-1}(\infty)$ be the locus of all poles of $f \in \mathbb{H}$ (inclusive of the cusp of $\mathrm{SL}_2(\mathbb{Z})$). The S- and T-segments at τ_0 are smooth paths in $\mathbb{H} \setminus \Gamma \cdot f^{-1}(\infty)$,

$$\gamma_{\tau_0}^S : -1/\tau_0 \longrightarrow \tau_0, \quad \gamma_{\tau_0}^T : \tau_0 - 1 \longrightarrow \tau_0, \quad (8)$$

running from $S^{-1}\tau_0 = -1/\tau_0$ and $T^{-1}\tau_0 = \tau_0 - 1$ to τ_0 .

Avoiding poles is automatic for $f \in M_k^!$, as long as the base-point τ_0 , is not at a cusp of $\mathrm{SL}_2(\mathbb{Z})$, i.e. there is only the trivial homotopy class. When $f \in F_k$, and f has poles at finite and non-zero imaginary heights, integration contours fall into nontrivial homotopy classes.

Definition 2.2 (*Hurwitz T-kernel*). Let $\zeta(s, x) = \sum_{m \geq 0} (m+x)^{-s}$ as a function $\zeta : \mathbb{C}_s \times (\mathbb{C} \setminus \mathbb{N})_x \rightarrow \mathbb{C}$ denote the Hurwitz zeta function (meromorphic in s , holomorphic in $x \in \mathbb{H}$). We define Hurwitz T-kernel as

$$\tilde{k}_T(\tau, s) := \zeta(1-s, \tau+1) - e^{i\pi(s-1)} \zeta(1-(k-s), \tau+1). \quad (9)$$

Definition 2.3 (*L-integral*). Let $f \in F_k$, $\tau_0 \in \mathbb{H}$, and $s \in \mathbb{C}$. Further, let $\gamma_{\tau_0}^S, \gamma_{\tau_0}^T$ be the S- and T-segments at τ_0 , as in Definition 2.1, and let $\tilde{k}_T$ be as defined in Definition 2.2. The L-integral of f is

$$L^*(f, s; \gamma_{\tau_0}^S, \gamma_{\tau_0}^T) := e^{-\pi i s/2} \left[\int_{\gamma_{\tau_0}^S} f(\tau) \tau^{s-1} d\tau + \int_{\gamma_{\tau_0}^T} f(\tau) \tilde{k}_T(\tau, s) d\tau \right]. \quad (10)$$

We use this *L-integral* to define a generalized *L-function*:

Definition 2.4 (*L-function*). Let $L^*(f, s; \gamma_{\tau_0}^S, \gamma_{\tau_0}^T)$ be as defined in 2.3. We define the generalized L-function of $f \in F_k$ to be,

$$L(f, s; \gamma_{\tau_0}^S, \gamma_{\tau_0}^T) := \frac{(2\pi)^s}{\Gamma(s)} L^*(f, s; \gamma_{\tau_0}^S, \gamma_{\tau_0}^T). \quad (11)$$

Having defined the L-integral and L-function, we prove they are manifestly finite for $f \in F_k$:

Lemma 2.5 (Finiteness). *For $f \in F_k$, both integrals in Definition 2.3 are finite for every $s \in \mathbb{C}$, every $\tau_0 \in \mathbb{H}$, and every choice of paths $\gamma_{\tau_0}^S, \gamma_{\tau_0}^T$.*

Proof. Both segments are compact and lie in $\mathbb{H} \setminus \Gamma \cdot f^{-1}(\infty)$, so f is continuous on them; τ^{s-1} is continuous on $\mathbb{H}$, and $\tilde{k}_T(\tau, s)$ is continuous there since $\zeta(\nu, \tau + 1)$ is holomorphic in $\tau \in \mathbb{H}$ (Definition 2.2). A continuous integrand on a compact path is integrable, for every $s \in \mathbb{C}$. $\square$

We now turn to path-dependence and τ_0 -independence of the L-integral and the L-function of $f \in F_k$. This holds for all $f \in M_k^!$:

Proposition 2.6 (Basepoint and path independence). *For $f \in M_k^!$ the value $L^*(f, s; \gamma_{\tau_0}^S, \gamma_{\tau_0}^T)$ is independent of $\tau_0 \in \mathbb{H}$ and of the paths: with no interior poles, $\mathbb{H}$ is simply connected, so all admissible contours are homotopic. We write $L^*(f, s)$ for the common value and $L(f, s) := \frac{(2\pi)^s}{\Gamma(s)} L^*(f, s)$.*

Proof. With no interior poles the integrands $f(\tau)\tau^{s-1}$ and $f(\tau)\tilde{k}_T(\tau, s)$ are holomorphic on $\mathbb{H}$, which is simply connected; by Cauchy's theorem each integral is unchanged under any deformation fixing its endpoints, hence depends on those endpoints alone. This is path-independence.

For independence of τ_0 we differentiate the two integrals and watch them cancel. The T -segment runs from $T^{-1}\tau_0 = \tau_0 - 1$ to τ_0 — two endpoints a unit apart, moving together with τ_0 — so by Leibniz's rule and the T -periodicity $f(\tau_0 - 1) = f(\tau_0)$,

$$\partial_{\tau_0} \int_{\gamma^T} f \tilde{k}_T = f(\tau_0) \tilde{k}_T(\tau_0, s) - f(\tau_0 - 1) \tilde{k}_T(\tau_0 - 1, s) = f(\tau_0) \delta_T \tilde{k}_T(\tau_0 - 1, s), \quad (12)$$

where $\delta_T \tilde{k}_T(\tau, s) := \tilde{k}_T(\tau + 1, s) - \tilde{k}_T(\tau, s)$. This is the whole role of δ_T : the unit gap between the endpoints turns the ∂_{τ_0} boundary term into a unit shift of the kernel. That shift is elementary — the kernel was built for it — since $\zeta(\nu, x + 1) - \zeta(\nu, x) = -x^{-\nu}$ gives, on both terms at $x = \tau + 1$,

$$\delta_T \tilde{k}_T(\tau, s) = -(\tau + 1)^{s-1} + e^{i\pi(s-1)}(\tau + 1)^{k-1-s} \quad (13)$$

$$\implies \partial_{\tau_0} \int_{\gamma^T} f \tilde{k}_T = f(\tau_0) [-\tau_0^{s-1} + e^{i\pi(s-1)} \tau_0^{k-1-s}]. \quad (14)$$

The S -segment runs $-1/\tau_0 \rightarrow \tau_0$; its lower endpoint moves with $\partial_{\tau_0}(-1/\tau_0) = \tau_0^{-2}$, so by Leibniz, the modularity $f(-1/\tau_0) = \tau_0^k f(\tau_0)$, and the phase $(-1/\tau_0)^{s-1} = e^{i\pi(s-1)} \tau_0^{1-s}$,

$$\partial_{\tau_0} \int_{\gamma^S} f \tau^{s-1} = f(\tau_0) \tau_0^{s-1} - f(-1/\tau_0) (-1/\tau_0)^{s-1} \tau_0^{-2} = f(\tau_0) [\tau_0^{s-1} - e^{i\pi(s-1)} \tau_0^{k-1-s}]. \quad (15)$$

The two brackets are negatives of each other, so $\partial_{\tau_0} L^* = e^{-\pi i s/2} (\partial_{\tau_0} \int_{\gamma^S} + \partial_{\tau_0} \int_{\gamma^T}) = 0$. $\square$

Remark 2.7 (Classical L-function contour). *Let $f \in S_k$. As $\tau_0 \rightarrow i\infty$, the S -segment maps onto the standard L-function contour, $[0, i\infty]$. Simultaneously, as $f(it) \sim e^{-2\pi t}$ as $t \rightarrow \infty$, the T -segment vanishes exponentially. Thus, in this limit, $L^*(f, s)$ exactly maps onto the classical L-function integral [4]:*

$$\lim_{\tau_0 \rightarrow i\infty} L^*(f, s) = \int_0^\infty t^{s-1} f(it) dt. \quad (16)$$

3 L -functions for weak modular forms

Proposition 3.1 (Hurwitz-Gamma). *Let $\Gamma(s, x) := \int_x^\infty e^{-t} t^{s-1} dt$ denote the incomplete Gamma function. For $n \neq 0$, $\nu \in \mathbb{C}$, and $z \in \mathbb{H}$,*

$$\int_z^{z+1} e^{2\pi i n u} \zeta(\nu, u) du = e^{i\pi(1-\nu)/2} (2\pi n)^{\nu-1} \Gamma(1-\nu, -2\pi i n z). \quad (17)$$

At $z = i$ the argument $-2\pi i n z$ collapses to $2\pi n$, giving $\Gamma(1-\nu, 2\pi n)$.

Proof. Write $\zeta(\nu, u) = \sum_{m \geq 0} (m+u)^{-\nu}$ and shift $u \mapsto u+m$ in the m -th term; as $e^{2\pi i n(u+m)} = e^{2\pi i n u}$ for integer m , the unit segments stack into one horizontal ray,

$$\int_z^{z+1} e^{2\pi i n u} \zeta(\nu, u) du = \sum_{m \geq 0} \int_{z+m}^{z+1+m} e^{2\pi i n v} v^{-\nu} dv = \int_z^{z+\infty} e^{2\pi i n v} v^{-\nu} dv. \quad (18)$$

Substitute $w = -2\pi i n v$, so that $v = iw/(2\pi n)$, $dv = i dw/(2\pi n)$, and $e^{2\pi i n v} = e^{-w}$: the lower endpoint becomes $w = -2\pi i n z$, and for $n > 0$ the ray rotates to $w \rightarrow +\infty$ along the positive reals — legal, since $e^{-w} w^{-\nu}$ decays in the right half-plane, so the closing arc vanishes. With $v^{-\nu} = i^{-\nu} (2\pi n)^{\nu} w^{-\nu}$,

$$\int_z^{z+\infty} e^{2\pi i n v} v^{-\nu} dv = i^{1-\nu} (2\pi n)^{\nu-1} \int_{-2\pi i n z}^{\infty} e^{-w} w^{-\nu} dw = e^{i\pi(1-\nu)/2} (2\pi n)^{\nu-1} \Gamma(1-\nu, -2\pi i n z). \quad (19)$$

For $n < 0$ the ray instead grows as $\text{im } v \rightarrow \infty$ and the rotation is illegal; but both sides are analytic in n and agree for $n > 0$, so the identity continues to $n < 0$ with Γ on its principal branch. At $n = 0$ the prefactor $(2\pi n)^{\nu-1}$ is singular — that mode, the constant Fourier coefficient, is handled separately. $\square$

Theorem 3.2 (L -function for weakly holomorphic forms). *For $f \in M_k^!$, we may set $\tau_0 = i$, and prove*

$$L^*(f, s) = -c_f(0) \left(\frac{1}{s} + \frac{i^k}{k-s} \right) + \sum_{n \neq 0} c_f(n) \left(\frac{\Gamma(s, 2\pi n)}{(2\pi n)^s} + i^k \frac{\Gamma(k-s, 2\pi n)}{(2\pi n)^{k-s}} \right). \quad (20)$$

This L -integral satisfies the functional equation $L^(f, s) = i^k L^*(f, k-s)$.*

Proof. By path independence and base-point independence from Proposition 2.6, we may freely set $\tau_0 = i$. Here, the S -segment degenerates ($-1/i = i$), and the L -integral simplifies dramatically:

$$L^*(f, s) = e^{-\pi i s/2} \int_{i-1}^i f(\tau) \tilde{k}_T(\tau, s) d\tau. \quad (21)$$

We then expand f in terms of its q -series, $\sum_n c_f(n) e^{2\pi i n \tau}$, and substitute $u = \tau + 1$, and note that $e^{2\pi i n(u-1)} = e^{2\pi i n u}$. Here, each $n \neq 0$ mode is handled by a straightforward substitution of ν, z into Proposition 3.1: Substituting $1-\nu = s$ gives the first term, and substituting $1-\nu = k-s$ gives the second:

$$e^{-\pi i s/2} \left(e^{\pi i s/2} \frac{\Gamma(s, 2\pi n)}{(2\pi n)^s} - e^{i\pi(s-1)} e^{\pi i(k-s)/2} \frac{\Gamma(k-s, 2\pi n)}{(2\pi n)^{k-s}} \right) = \frac{\Gamma(s, 2\pi n)}{(2\pi n)^s} + i^k \frac{\Gamma(k-s, 2\pi n)}{(2\pi n)^{k-s}}. \quad (22)$$

So each $n \neq 0$ mode contributes $\Gamma(s, 2\pi n)/(2\pi n)^s + i^k \Gamma(k-s, 2\pi n)/(2\pi n)^{k-s}$.

Crucially, we may evaluate the $n = 0$ mode's contribution as follows. The Hurwitz zeta function, $\zeta(\sigma, a) = \sum_{m \geq 0} (m + a)^{-\sigma}$, satisfies

$$\int \zeta(\sigma, a) da = -\frac{\zeta(\sigma - 1, a)}{\sigma - 1} + C \quad (23)$$

for $\sigma \neq 1$. Now, note that the finite difference $\zeta(\sigma - 1, a + 1) - \zeta(\sigma - 1, a) = -a^{1-\sigma}$ follows directly from the sum-representation of $\zeta(\sigma, a)$, when evaluated at $a = i$ gives a closed form solution for the definite integral,

$$\int_{i-1}^i \zeta(\sigma, \tau + 1) d\tau = \int_i^{i+1} \zeta(\sigma, u) du = \frac{i^{1-\sigma}}{\sigma - 1}. \quad (24)$$

Finally, applying this to the two distinct terms of $\tilde{k}_T(\tau, s)$ at $\sigma = 1 - s$ and $\sigma = 1 - (k - s)$ gives

$$e^{-\pi i s/2} \int_{i-1}^i c_f(0) \tilde{k}_T(\tau, s) d\tau = -c_f(0) \left(\frac{1}{s} + \frac{i^k}{k - s} \right). \quad (25)$$

Combining the $n = 0$ contributions with the $n \neq 0$ contributions reproduces Eq. (20). Finally, the functional equation, $L^*(f, s) = i^k L^*(f, k - s)$, follows by direct inspection. $\square$

3.1 Explicit S- and T-contours for general τ_0

Lemma 3.3 (*T-segment closed form on $M_k^!$*). For $f \in M_k^!$, for generic $\tau_0 \in \mathbb{H}$ away from cusps, denote the T-contour's contribution to $L^*(f, s)$ as $L_T^*(f, s; \gamma_{\tau_0}^T)$. For generic τ_0 , we have:

$$L_T^*(f, s; \gamma_{\tau_0}^T) = -c_f(0) \left(\frac{(\tau_0/i)^s}{s} + \frac{i^k (\tau_0/i)^{k-s}}{k-s} \right) + \sum_{n \neq 0} c_f(n) \left(\frac{\Gamma(s, -2\pi i n \tau_0)}{(2\pi n)^s} + i^k \frac{\Gamma(k-s, -2\pi i n \tau_0)}{(2\pi n)^{k-s}} \right). \quad (26)$$

At $\tau_0 = i$ the constant mode collapses to $-c_f(0)(1/s + i^k/(k-s))$, recovering Theorem 3.2.

Proof. Expand $f = \sum_n c_f(n) q^n$ on the compact segment $[\tau_0 - 1, \tau_0]$, and restrict attention to the sub-sum, $\sum_{n \neq 0}$. For each such $n \neq 0$, apply the Proposition 3.1 in $\tilde{k}_T(\tau, s) = \zeta(1-s, \tau+1) - e^{i\pi(s-1)} \zeta(s-(k-1), \tau+1)$ at $\nu = 1-s$ and $\nu = s-(k-1)$. The phases for the second term work-out identically as in the proof of Theorem 3.2.

For $n = 0$, the constant $c_f(0)$ multiplies $e^{-\pi i s/2} \int_{\tau_0-1}^{\tau_0} \tilde{k}_T(\tau, s) d\tau$; the Hurwitz antiderivative $\int_{\tau_0-1}^{\tau_0} \zeta(\sigma, \tau+1) d\tau = \tau_0^{1-\sigma}/(\sigma-1)$ — from $\partial_\tau \zeta(\sigma-1, \tau+1) = -(\sigma-1)\zeta(\sigma, \tau+1)$ and the shift $\zeta(\sigma-1, \tau_0+1) - \zeta(\sigma-1, \tau_0) = -\tau_0^{1-\sigma}$ — applied to the two halves of $\tilde{k}_T$ at $\sigma = 1-s$ and $\sigma = s-k+1$ yields $-e^{-\pi i s/2} \tau_0^s/s$ and $-e^{\pi i s/2} \tau_0^{k-s}/(k-s)$ respectively. $\square$

Corollary 3.4 (*T-segment cusp asymptotics*). Let $f \in S_k^!$, where $f = \sum_{n \geq -N} c_f(n) q^n$ for some $-N \in \mathbb{Z}$. The T-contour's contribution to $L^*(f, s)$ is given by Lemma 3.3. As $\tau_0 = it$ goes to the cusp $i\infty$, the T-contour scales exactly as $f(it)$ scales: $L_T^*(f, s; \gamma_{\tau_0}^T) \sim f(it) \sim e^{2\pi N t}$.

Proof. For the asymptotics, $\Gamma(\sigma, z) \sim z^{\sigma-1} e^{-z}$ as $|z| \rightarrow \infty$ for $|\arg z| < \frac{3\pi}{2}$ [10, §8.11] gives $|c_f(n)|$ -weighted go as $e^{-2\pi n \operatorname{im}(\tau_0)}$. The smallest mode n in the q-series, i.e. $n = -N$, dominates the scaling of $f(it)$ and of $L_T^*(f, s; \gamma_{it}^T)$. Combining these results yields the claimed asymptotics. $\square$

Now, exploiting τ_0 -independence of $L^*(f, s)$, its explicit value as given in Theorem 3.2, and the explicit T-contribution $L_T^*(f, s; \gamma_{\tau_0}^T)$ trivially implies clean, closed-form expressions for the S-contour's contribution to $L^*(f, s)$:

Corollary 3.5 (*S*-segment closed form). *For $f \in M_k^!$ and $\tau_0 \in \mathbb{H}$, the S -contribution $L_S^*(f, s; \gamma_{\tau_0}^S) := e^{-\pi i s/2} \int_{-1/\tau_0}^{\tau_0} f(\tau) \tau^{s-1} d\tau$ is the full L -value minus the T -contribution at the same base point, hence an explicit difference of incomplete gammas between the base points i and τ_0 :*

$$L_S^*(f, s; \gamma_{\tau_0}^S) = L^*(f, s) - L_T^*(f, s; \gamma_{\tau_0}^T) \quad (27)$$

$$\begin{aligned} &= c_f(0) \left(\frac{e^{-\pi i s/2} \tau_0^s - 1}{s} + \frac{e^{\pi i s/2} \tau_0^{k-s} - i^k}{k-s} \right) \\ &+ \sum_{n \neq 0} c_f(n) \left(\frac{\Gamma(s, 2\pi n) - \Gamma(s, -2\pi i n \tau_0)}{(2\pi n)^s} + i^k \frac{\Gamma(k-s, 2\pi n) - \Gamma(k-s, -2\pi i n \tau_0)}{(2\pi n)^{k-s}} \right). \end{aligned} \quad (28)$$

Proof. Line (27) is the definition $L^*(f, s; \gamma_{\tau_0}^S, \gamma_{\tau_0}^T) = L_S^*(\gamma_{\tau_0}^S) + L_T^*(\gamma_{\tau_0}^T)$ together with the τ_0 -independence of L^* (Proposition 2.6); evaluating at $\tau_0 = i$, where $L_S^*(\gamma_i^S) = 0$, identifies $L^*(f, s) = L_T^*(\gamma_i^T)$. Line (28) then inserts the closed forms of Theorem 3.2 for $L^*(f, s)$ and Lemma 3.3 for $L_T^*(\gamma_{\tau_0}^T)$: the $n \neq 0$ modes subtract base point by base point into the incomplete-gamma differences, and the two constant modes subtract to the $c_f(0)$ -terms in Eq. (28). $\square$

3.2 Classical Mellin recovery and canonical uniqueness.

As discussed in Remark 2.7, for $f \in S_k$, taking the cusp-limit $\tau_0 \rightarrow i\infty$ in Theorem 3.2 collapses the S -segment onto the classical Mellin contour $[0, i\infty]$ while the T -segment vanishes exponentially, since $f(it) = O(e^{-2\pi t})$ on the segment as $\tau = it \rightarrow i\infty$. I.e., here, one recovers the classical L -integral of f without obstruction:

$$\lim_{\tau_0 \rightarrow i\infty} L^*(f, s; \gamma_{\tau_0}^S, \gamma_{\tau_0}^T) = \int_0^\infty t^{s-1} f(it) dt. \quad (29)$$

Of course, path- and base-point independence of $L^*(f, s) = L^*(f, s; \gamma_{\tau_0}^S, \gamma_{\tau_0}^T)$ is guaranteed by Theorem 3.2 for all $f \in M_k^!$. However, these L -integrals are the sums of the S -contour and T -contour. Finiteness is ensured only of the sum of these two parts. Both contours may, and indeed do, grow without bound as $\tau_0 \rightarrow i\infty$ when $f \notin M_k$, as is highlighted in Corollary 3.4:

As $\tau_0 \rightarrow i\infty$ the polar mode q^{-N} of a properly weakly holomorphic modular form $f \in M_k^!$ makes the T -contribution grow like $e^{2\pi N \operatorname{Im} \tau_0}$ and the S -contribution diverge oppositely. Yet their sum, the τ_0 -independent value $L^*(f, s)$, remains finite and constant over all finite $\tau_0 \in \mathbb{H}$. Discarding $\gamma_{\tau_0}^T$ therefore does violence to the result: When the S -contour is isolated from its T -partner, and $\tau_0 \rightarrow i\infty$, it diverges. This is the precise reason that [1] needed to regularize their divergent L -integral representation for properly weak cusp forms.

Remark 3.6. *In textbook treatments [4], when $f \in M_k$, its L -integral is defined as*

$$\begin{aligned} L^*(f, s)|_{\text{cl}} &:= \int_0^\infty (f(it) - c_f(0)) t^{s-1} dt = \frac{\Gamma(s)}{(2\pi)^s} \sum_{n=1}^\infty \frac{c_f(n)}{n^s} \\ &= -c_f(0) \left(\frac{1}{s} + \frac{i^k}{k-s} \right) + \sum_{n=1}^\infty c_f(n) \left(\frac{\Gamma(s, 2\pi n)}{(2\pi n)^s} + i^k \frac{\Gamma(k-s, 2\pi n)}{(2\pi n)^{k-s}} \right) \end{aligned} \quad (30)$$

Classically, this constant-term subtraction is needed because $\int_0^\infty t^{s-1} dt$ is a pure divergence, so the q -series cannot commute with the integration contour.

However, as is clear from our general Theorem 3.2 for L -integrals for $f \in M_k^!$, and from the specific results of the S - and T -contours in Lemma 3.3 and Corollary 3.5, our L -integral requires no

such explicit subtraction of the constant term $-c_f(0)$. Thus, not only does our prescription naturally handle divergences at poles when $f \in M_k^!$ has polar terms in its q -series, but it also obviates the need to by-hand remove the constant mode from the kernel of the L -integral that is needed when defining L -functions in terms of the classical Mellin contour, $\int_0^\infty$.

3.3 Period polynomials, periods and quasi-periods

Because $L^*(f, s)$ is canonical for every $f \in M_k^!$, the period polynomial needs no separate construction — it is the generating polynomial of the critical values.

Definition 3.7 (Period polynomial). *For $f \in S_k^!$ and $n = k - 2$, the period polynomial is*

$$r_f(X, Y) := (2\pi i)^{n+1} \sum_{\ell=0}^n (-1)^\ell \binom{n}{\ell} i^{\ell+1} L^*(f, \ell+1) X^{n-\ell} Y^\ell. \quad (31)$$

Proposition 3.8 (Period polynomial as a contour integral). *With $(X - \tau Y)^n$ in place of the L -weight τ^{s-1} , and $\mathbf{K}_T(\tau; X, Y) := \sum_{\ell=0}^n \binom{n}{\ell} (-1)^\ell \tilde{k}_T(\tau, \ell+1) X^{n-\ell} Y^\ell$ in place of $\tilde{k}_T$, the same two segments assemble r_f directly:*

$$r_f(X, Y) = (2\pi i)^{n+1} \left[\int_{\gamma_{\tau_0}^S} f(\tau) (X - \tau Y)^n d\tau + \int_{\gamma_{\tau_0}^T} f(\tau) \mathbf{K}_T(\tau; X, Y) d\tau \right]. \quad (32)$$

Proof. By Definition 2.3, the $X^{n-\ell} Y^\ell$ coefficient of the bracket is

$$r_f|_{X^{n-\ell} Y^\ell} = (-1)^\ell \binom{n}{\ell} \left[\int_{\gamma^S} f(\tau) \tau^\ell + \int_{\gamma^T} f(\tau) \tilde{k}_T(\tau, \ell+1) \right] = (-1)^\ell \binom{n}{\ell} i^{\ell+1} L^*(f, \ell+1). \quad (33)$$

This matches definition 3.7 term by term, and proves the claim. $\square$

Remark 3.9 (The role of the T -kernel). *The Hurwitz T -kernel in this note was originally uncovered in order to reproduce the manifestly finite computations of the quasi-periods for $\hat{\Delta} = 1/q + O(q^2) \in S_{12}^!$ originally given in [2]. Its role in the period polynomial (32) is sharp: the S -segment alone is not invariant under the parabolic shift $\tau_0 \mapsto \tau_0 + 1$, and the T -segment $\int_{\gamma_{\tau_0}^T} f(\tau) \mathbf{K}_T(\tau; X, Y) d\tau$ is precisely the term that cancels this variation, leaving $r_f(X, Y)$ independent of the basepoint τ_0 (Proposition 2.6). This basepoint-independence is the finite-contour counterpart of the parabolicity condition in Eichler–Shimura theory.*

Quasi-periods and their determinant: It is interesting to collect the quasi-periods computable from our L -integrals/period-polynomials for $f \in S_k^!$ (when $\dim(S_k) = 1$).

To this effect, we recall the notation of [5]. Decomposing the modular weight $k = 12\lambda + k'$ where $k' \in \{0, 4, 6, 8, 10, 14\}$, we see $\lambda = \dim(S_k)$ when $\lambda \geq 0$. For any λ , we may denote $f_{k,m} := 1/q^m + O(q^{\lambda+1}) \in M_k^!$. When $\dim(S_k) = 1$, we have $f_{k,m} = 1/q^m + O(q^2)$. Here, define $\Delta_k := f_{k,-1} = q + O(q^2) \in S_k$ and $\hat{\Delta}_k := f_{k,1} = 1/q + O(q^2) \in S_k^!$. Further, define

$$\text{pd}_k := \begin{pmatrix} \omega^+(\Delta_k) & \omega^-(\Delta_k) \\ \eta^+(\hat{\Delta}_k) & \eta^-(\hat{\Delta}_k) \end{pmatrix}, \quad \alpha_k := \frac{(-1)^{1+k/2} \det(\text{pd}_k)}{(2\pi i)^{k-1} (k-2)!} \quad (34)$$

Brown–Hain [7, Prop. 5.6] show that $\det(\text{pd}_k)$ satisfies a Legendre-type relation, $\det(\text{pd}_k) \in \mathbb{Q}^\times \cdot (2\pi i)^{k-1}$. Brown [2, §8.3] evaluates the constant at $k = 12$; we reproduce this result at $k = 12$ and give the analogous rational multiples for $k \in \{16, 18, 20, 22, 26\}$ in Table 1, below:

k	$\omega^+(f_k)$	$\omega^-(f_k)/i$	$\eta^+(\hat{\Delta}_k)$	$\eta^-(\hat{\Delta}_k)/i$	α_k
12	-6.89168e7	-5.58502e6	1.27202e11	1.02767e10	1
16	-8.81481e11	-1.54063e10	4.14120e13	6.83541e11	13/30
18	-5.07833e13	-5.43305e12	-2.19200e14	-3.36339e13	2/3
20	-4.22882e16	-3.73170e14	6.38640e16	-1.89823e14	17/5
22	-1.49974e18	-4.24489e16	1.43795e18	-4.41050e16	19/21
26	-1.18673e22	-3.60961e21	1.18653e22	-3.61016e21	23/15

Table 1: Periods $\omega^\pm(\Delta_k)$ and quasi-periods $\eta^\pm(\hat{\Delta}_k)$, and the rational number α_k . Note, while $\alpha_{12} = 1$ matching Brown [2], α_k for $k \geq 16$ appear to be new, interesting, rational numbers, albeit with implicit dependence on the period-polynomial basis normalizations. While this may obfuscate novelty, we have not found a clear pattern. Details of these extractions, and 50-digit extraction of these rational numbers is in [9].

Remark 3.10 (External check: the Petersson norm). *The periods $r_m(f) = \int_0^{i\infty} \tau^m f(\tau) d\tau = i^{m+1} L^*(f, m+1)$ read off the finite-contour integral are the classical critical periods, so they must obey Haberland’s formula*

$$-6(-2i)^{k-2} \langle f, f \rangle = \sum_{m+n \leq k-2} \binom{k-2}{m+n} \binom{m+n}{m} (-1)^m \operatorname{im}(r_m(f) \overline{r_n(f)}), \quad (35)$$

which recovers the Petersson norm $\langle f, f \rangle$ from the period polynomial [8, Thm. 5.2]. Unlike α_k , this test is independent of the period-polynomial basis normalization. For Δ ($k = 12$) our L^* -values reproduce Zagier’s $\langle \Delta, \Delta \rangle = 1.03536 \dots \times 10^{-6}$ to 14 digits; at $k = 16$ they match the fundamental-domain integral $\int_{\mathcal{F}} |f|^2 y^{k-2} dx dy = 2.16906 \dots \times 10^{-6}$ to 16 digits.

4 Meromorphic modular forms

For $f \in F_k$ with poles in $\mathbb{H}$, the integrals of Definition 2.3 depend on the contours. Write the L -integral as the sum of its two segments,

$$L^*(f, s; \gamma_{\tau_0}^S, \gamma_{\tau_0}^T) = L_S^*(f, s; \gamma_{\tau_0}^S) + L_T^*(f, s; \gamma_{\tau_0}^T), \quad \begin{cases} L_S^* := e^{-\pi i s/2} \int_{\gamma_{\tau_0}^S} f(\tau) \tau^{s-1} d\tau, \\ L_T^* := e^{-\pi i s/2} \int_{\gamma_{\tau_0}^T} f(\tau) \tilde{k}_T(\tau, s) d\tau, \end{cases} \quad (36)$$

each finite for every $s \in \mathbb{C}$ (Lemma 2.5). This section shows the value depends only on the homotopy class of the pair, quantifies the class dependence — each segment contributes its own residues — and proves an exponentially convergent series representation, valid in every class and at every $s \in \mathbb{C}$.

For $f \in F_k$, poles of finite order away from the cusp of $\mathrm{SL}_2(\mathbb{Z})$ make the contours matter: the L -integral is organized by the homotopy classes of $(\gamma_{\tau_0}^S, \gamma_{\tau_0}^T)$ in $\mathbb{H} \setminus \Gamma \cdot f^{-1}(\infty)$.

Proposition 4.1. *For $f \in F_k$, the L -integral $L^*(f, s; \gamma_{\tau_0}^S, \gamma_{\tau_0}^T)$ depends only on the homotopy class of its contours, and not on the basepoint τ_0 . At fixed τ_0 each of L_S^* , L_T^* is separately path-independent; a change of basepoint moves value between the two, leaving only their sum invariant.*

Proof. At fixed τ_0 the integrands $f \tau^{s-1}$ and $f \tilde{k}_T$ are holomorphic on $\mathbb{H} \setminus \Gamma \cdot f^{-1}(\infty)$, so Cauchy’s theorem makes each summand of (36) constant on the homotopy class of its own segment. Under

a motion of the basepoint, the derivative $\partial_{\tau_0} L^*$ is evaluated at the moving endpoints alone, and the computation of Proposition 2.6 — which uses only $f(\tau_0 - 1) = f(\tau_0)$ and $f(-1/\tau_0) = \tau_0^k f(\tau_0)$ — cancels the boundary terms of L_S^* against those of L_T^* ; the interior poles lie off the contours and never enter. Hence $\partial_{\tau_0} L^* = 0$ along any motion through $\mathbb{H} \setminus \Gamma \cdot f^{-1}(\infty)$, while each summand separately varies. $\square$

Definition 4.2. Fix $f \in F_k$ and $\tau_0 \in \mathbb{H}$, and let $[\gamma_{\tau_0}^S], [\gamma_{\tau_0}^T]$ denote homotopy classes of the two segments in $\mathbb{H} \setminus \Gamma \cdot f^{-1}(\infty)$, endpoints fixed. By Proposition 4.1 the assignment

$$L^*(f, s; [\gamma_{\tau_0}^S], [\gamma_{\tau_0}^T]) := L_S^*(f, s; \gamma_{\tau_0}^S) + L_T^*(f, s; \gamma_{\tau_0}^T), \quad \text{any representatives,} \quad (37)$$

is well-defined; by the same proposition, motion of the basepoint changes the two classes jointly but not the value.

Poles strictly above the line $\text{im } \tau = 1$ force exponential growth on q -series coefficients — $|c_f(n)| \gtrsim n^{m-1} e^{2\pi n \text{im } \tau_p}$ for a pole of order m at τ_p [3] — and defeat term-by-term integration along γ^T . We remove exactly these poles, with the projection of [3].

Definition 4.3. Let $f \in F_k$ have interior poles at the points τ_p of a set $\mathcal{P}_f$ of Γ -inequivalent representatives, and write $\text{Li}_s(z) = \sum_{n>0} z^n n^{-s}$. Since $\text{Li}_{1-m}(e^{2\pi i(\tau - \tau_p)})$ has a pole of order m at τ_p , the principal part of f there is matched by a unique polylog projection

$$\hat{P}_{\tau_p} f(\tau) := \sum_{m>0} r_{f, \tau_p}^*(m) \text{Li}_{1-m}(e^{2\pi i(\tau - \tau_p)}), \quad (38)$$

its coefficients $r_{f, \tau_p}^*(m)$ being the principal-part data of f at τ_p . The projection $\hat{P} := \sum_{\text{im } \tau_p > 1} \hat{P}_{\tau_p}$ removes the principal part of every pole strictly above the line $\text{im } \tau = 1$.

Definition 4.4. With $\hat{P}$ as in Definition 4.3, $c_{Rf}(n)$ denotes the Fourier coefficients of $(I - \hat{P})f$ in its expansion valid for large $\text{im } \tau$:

$$(I - \hat{P})f = \sum_n c_{Rf}(n) q^n. \quad (39)$$

Definition 4.5. $F(\infty)$ is the space of meromorphic g on $\mathbb{H}$ with $g(\tau) = g(\tau + 1)$, finitely many poles in any compact subset of the strip $\{\text{Im } \tau > 0, |\text{Re } \tau| \leq \frac{1}{2}\}$, and $g(\tau) = O(e^{C \text{Im } \tau})$ as $\text{Im } \tau \rightarrow \infty$ for some C .

Then $F_k \subset F(\infty)$, and $(I - \hat{P})f \in F(\infty)$ — the polylog tails are T -periodic but not modular, so the projection leaves F_k .

We further define, for $n \neq 0$, the single-mode contribution of Theorem 3.2,

$$G_{s,k}(n) := \frac{\Gamma(s, 2\pi n)}{(2\pi n)^s} + i^k \frac{\Gamma(k-s, 2\pi n)}{(2\pi n)^{k-s}}, \quad (40)$$

and, for one tail of Definition 4.3, the L -integral along the T -segment at $\tau_0 = i$, where γ^S degenerates ($-1/i = i$):

$$I_N(s, z) := e^{-\pi i s/2} \int_{\gamma_i^T} \text{Li}_{-N}(e^{2\pi i(\tau - z)}) \tilde{k}_T(\tau, s) d\tau. \quad (41)$$

Lemma 4.6. For $\text{Im } z > 1$, polylog inversion $\text{Li}_{-N}(w) = (-1)^{N+1} \text{Li}_{-N}(1/w)$ resums (41) into

$$I_N(s, z) = \delta_{N,0} \left(\frac{1}{s} + \frac{i^k}{k-s} \right) + (-1)^{N+1} \sum_{n \geq 1} n^N e^{2\pi i n z} G_{s,k}(-n), \quad (42)$$

convergent exponentially with base $e^{-2\pi(\text{Im } z - 1)}$.

Proof. For $\text{Im } z > 1$ one has $|e^{2\pi i(\tau-z)}| > 1$ on $[i-1, i]$, so the defining series of Li_{-N} diverges there; the inversion rewrites it through $\text{Li}_{-N}(e^{2\pi i(z-\tau)}) = \sum_{n \geq 1} n^N e^{2\pi i n(z-\tau)}$, convergent since $|e^{2\pi i(z-\tau)}| < 1$. Integrating term by term against $\tilde{k}_T$, each $e^{-\pi i s/2} \int_{i-1}^i e^{-2\pi i n \tau} \tilde{k}_T d\tau$ is the mode $-n$ building block of Theorem 3.2, namely $G_{s,k}(-n)$; for $N = 0$ the leftover -1 in $\text{Li}_0(w) = -1 - \text{Li}_0(1/w)$ integrates to $\frac{1}{s} + \frac{i^k}{k-s}$. The rate follows from $\Gamma(s, -2\pi n) \sim (2\pi n)^{s-1} e^{2\pi n}$ against $|e^{2\pi i n z}| = e^{-2\pi n \text{Im } z}$. $\square$

The series of Lemma 4.6 extends to the edge $z = i + x$, $x \in [-\frac{1}{2}, \frac{1}{2})$. Write $w := e^{2\pi i(z-i)}$, $Q := e^{2\pi i z}$, and $(a)_{(j)} := a(a-1)\cdots(a-j+1)$, $(a)_{(0)} := 1$; principal branches throughout.

Lemma 4.7. *As identities of meromorphic functions of $s \in \mathbb{C}$, for $\text{Im } z > 1$ and $N \geq 0$,*

$$I_N(s, z) = \delta_{N,0} \left(\frac{1}{s} + \frac{i^k}{k-s} \right) + (-1)^{N+1} \left[\Pi_N(s; w) + \Lambda_N(s; Q) + \mathcal{E}_N(s, z) \right], \quad (43)$$

where

$$\Pi_N(s; w) := \sum_{j=0}^{N+1} \frac{(-1)^{j+1} [(s-1)_{(j)} + i^k (k-s-1)_{(j)}]}{(2\pi)^{j+1}} \text{Li}_{j+1-N}(w), \quad (44)$$

$$\Lambda_N(s; Q) := \frac{\Gamma(s)}{(-2\pi)^s} \text{Li}_{s-N}(Q) + i^k \frac{\Gamma(k-s)}{(-2\pi)^{k-s}} \text{Li}_{k-s-N}(Q), \quad (45)$$

$$\begin{aligned} \mathcal{E}_N(s, z) := & - \sum_{n \geq 1} n^N Q^n \left[(s-1)_{(N+2)} \frac{\Gamma(s-N-2) - \Gamma(s-N-2, -2\pi n)}{(-2\pi n)^s} \right. \\ & \left. + i^k (k-s-1)_{(N+2)} \frac{\Gamma(k-s-N-2) - \Gamma(k-s-N-2, -2\pi n)}{(-2\pi n)^{k-s}} \right]. \end{aligned} \quad (46)$$

The terms of $\mathcal{E}_N$ are $O(n^{-3}|w|^n)$, uniformly on $\text{Im } z \geq 1$; hence (43) extends $I_N(s, \cdot)$ continuously to the edge away from $w = 1$, for every s and every N . At $w = 1$, i.e. $z \in i + \mathbb{Z}$, the obstruction is confined to the $\text{Li}_{\leq 1}$ terms of Π_N ; for $N = 0$ the Li_1 coefficient is $(1+i^k)/(2\pi)$, so for $k \equiv 2 \pmod{4}$ the extension is finite at $z = i$ as well.

Proof. $\square$

Remark 4.8. At integer $s \in \{1, \dots, k-1\}$ the Pochhammer symbols terminate all three layers and (43) closes in polylogarithms of w : e.g. at $k = 6$, $I_0(2, z) = \frac{1}{4} + \frac{\text{Li}_2(w)}{2\pi^2} - \frac{3\text{Li}_3(w)}{4\pi^3} + \frac{3\text{Li}_4(w)}{8\pi^4}$, so $I_0(2, i) = \frac{27}{80} - \frac{3\zeta(3)}{4\pi^3}$. At the central point the kernel itself vanishes for $k \equiv 2 \pmod{4}$ — $\tilde{k}_T(\tau, k/2) = [1 - e^{i\pi(k/2-1)}] \zeta(1 - \frac{k}{2}, \tau+1) \equiv 0$ — so there $L^* = L_S^*$; for Δ/E_6 the two γ^S -chambers at $s = 3$ carry $\pm\pi \text{Res}_{\tau=i}(\Delta/E_6)$ with principal value 0: the central L -value of a $k \equiv 2$ block is half a wall jump of Lemma 4.11.

The excluded point $z = i$ is reached through the wall: anchor the contour at $\tau_0 = i(1 + \delta)$ and let $\delta \rightarrow 0^+$. Write $H_N := \sum_{r=1}^N \frac{1}{r}$ and $c_j^+(s) := (s-1)_{(j)} + i^k (k-s-1)_{(j)}$.

Lemma 4.9. *Let $f \in F_k$ have a pole at i of order P , with principal part $\sum_{m=1}^P a_{-m}(\tau-i)^{-m}$, and let $\hat{P}_i f$ be its polylog projection. In the class of $\tau_0 = i(1 + \delta)$, the γ^S -passage through i taken as a principal value,*

$$\lim_{\delta \rightarrow 0^+} L^*(\hat{P}_i f, s) = \sum_{m=1}^P \frac{(-2\pi i)^m a_{-m}}{(m-1)!} \text{FP}_{m-1}(s, k), \quad (47)$$

$$\text{FP}_N(s, k) = \frac{c_N^+(s) H_N}{(2\pi)^{N+1}} + \sum_{\substack{j \geq 0 \\ j \neq N}} \frac{c_j^+(s) \zeta(j+1-N)}{(2\pi)^{j+1}} + \mathcal{E}_N^+(s) + \frac{(-1)^N N!}{(2\pi)^{N+1}} \sum_{t=0}^{N-1} \frac{(-1)^t (s-1)_{(t)}}{t! (N-t)} , \quad (48)$$

the j -sum taken to any $J \geq N+2$, with remainder

$$\mathcal{E}_N^+(s) = \sum_{n \geq 1} n^N e^{2\pi n} \left[(s-1)_{(J)} \frac{\Gamma(s-J, 2\pi n)}{(2\pi n)^s} + i^k (k-s-1)_{(J)} \frac{\Gamma(k-s-J, 2\pi n)}{(2\pi n)^{k-s}} \right]. \quad (49)$$

The individually divergent δ -powers and logarithms cancel by modularity alone; for a double pole the cancellation condition is exactly the S -fixed-point constraint $a_{-1} = \frac{i(k-2)}{2} a_{-2}$.

Proof. Write $u := 2\pi i(\tau - i)$ and $g_N(\tau) := \text{Li}_{-N}(e^u)$. The expansion $\text{Li}_\sigma(e^{-\mu}) = \Gamma(1-\sigma) \mu^{\sigma-1} + \sum_{r \geq 0} \zeta(\sigma-r) \frac{(-\mu)^r}{r!}$ at $\mu = -u$ gives

$$g_N(u) = \frac{(-1)^{N+1} N!}{u^{N+1}} + \sum_{p \geq 0} \frac{\zeta(-N-p)}{p!} u^p, \quad (50)$$

a single negative power per tail; matching principal parts is therefore diagonal and gives $\hat{P}_i f = \sum_{m=1}^P r_m g_{m-1}$ with $r_m = (-2\pi i)^m a_{-m}/(m-1)!$. Split (36) at $\tau_0 = i(1+\delta)$ and set $v := e^{-2\pi\delta}$.

T-segment. The poles of g_N lie on $i + \mathbb{Z}$, below $\gamma_{\tau_0}^T$, so $g_N = \sum_{n \geq 1} n^N e^{2\pi n} e^{2\pi i n \tau}$ converges uniformly on the segment and integrates mode by mode. For $F_n(\tau_0) := e^{-\pi i s/2} \int_{\tau_0-1}^{\tau_0} e^{2\pi i n \tau} \tilde{k}_T(\tau, s) d\tau$, periodicity of the mode and the difference equation of the kernel, $\tilde{k}_T(\tau_0, s) - \tilde{k}_T(\tau_0-1, s) = -\tau_0^{s-1} + e^{i\pi(s-1)} \tau_0^{k-s-1}$ (from $\zeta(\sigma, a) - \zeta(\sigma, a+1) = a^{-\sigma}$), determine $\partial_{\tau_0} F_n$; comparison with Theorem 3.2 at $\tau_0 = i$ then fixes

$$F_n(\tau_0) = \frac{\Gamma(s, -2\pi i n \tau_0)}{(2\pi n)^s} + i^k \frac{\Gamma(k-s, -2\pi i n \tau_0)}{(2\pi n)^{k-s}}, \quad (51)$$

principal branches, the argument $-2\pi i n \tau_0 = 2\pi n(1+\delta)$ positive. J -fold integration by parts, $\Gamma(s, x) = e^{-x} \sum_{j < J} (s-1)_{(j)} x^{s-1-j} + (s-1)_{(J)} \Gamma(s-J, x)$, resums the T -side of one tail into

$$T_N(\delta) = \sum_{j=0}^{J-1} \frac{(s-1)_{(j)} (1+\delta)^{s-1-j} + i^k (k-s-1)_{(j)} (1+\delta)^{k-s-1-j}}{(2\pi)^{j+1}} \text{Li}_{j+1-N}(v) + \mathcal{E}_N^+(s; \delta), \quad (52)$$

whose remainder has terms $O(n^{N-J-1} v^n)$ uniformly in $\delta \geq 0$ and equals (49) at $\delta = 0$. Expanding (52) with the same $\text{Li}_\sigma(e^{-\mu})$ formula, now at $\mu = 2\pi\delta$: each $j \neq N$ contributes $c_j^+(s) \zeta(j+1-N) (2\pi)^{-j-1}$ at order δ^0 ; for $j < N$ the singular powers $\Gamma(N-j) \mu^{j-N}$ meet the δ^{N-j} -coefficients of the $(1+\delta)$ -binomials, and the composition $(s-1)_{(j)} (s-1-j)_{(N-j)} = (s-1)_{(N)}$ collapses their total to $c_N^+(s) H_N (2\pi)^{-N-1}$; the term $j = N$ contributes only the logarithm $-c_N^+(s; \delta) \ln(2\pi\delta) (2\pi)^{-N-1}$.

S-segment. On the imaginary axis, $e^{-\pi i s/2} \int_{\gamma_S} g_N \tau^{s-1} d\tau = \frac{1}{2\pi} \int_{-2\pi\delta}^{2\pi\delta/(1+\delta)} \left(1 - \frac{u}{2\pi}\right)^{s-1} g_N(u) du$.

The analytic part of (50) integrates to $O(\delta)$. In the pole part, expand $(1 - \frac{u}{2\pi})^{s-1} = \sum_t \frac{(-1)^t (s-1)_{(t)}}{t! (2\pi)^t} u^t$: each power u^{t-N-1} with $t < N$ is an exact derivative, so its integral is a difference of endpoint antiderivatives — independent of the indentation at $u = 0$ — with δ^0 -coefficient $(2\pi)^{t-N}/(t-N)$; the power $t = N$ is the principal value $\ln(1+\delta) = O(\delta)$; $t > N$ is $O(\delta)$. Collecting gives the last term of (48). (A side-indentation at $u = 0$ differs from the principal value by half the wall jump of Lemma 4.11.)

Cancellation. By Proposition 4.1, $L^*(f, s)$ is constant for $\delta > 0$ small, and $L^*((I - \hat{P}_i)f, s)$ is continuous at $\tau_0 = i$, its γ^S -integral vanishing as the segment contracts onto the regular point i . Hence $L^*(\hat{P}_i f, s) = L^*(f, s) - L^*((I - \hat{P}_i)f, s)$ converges as $\delta \rightarrow 0^+$, so the coefficients of the divergent powers δ^{-r} and of $\ln \delta$ in $\sum_m r_m (T_{m-1} + S_{m-1})$ vanish identically — in s as well, one relation per power of s — and the limit is the sum of the δ^0 -coefficients: (47) with (48). For $P = 2$ the vanishing of the logarithm reads $r_1 c_0^+(s)(2\pi)^{-1} + r_2 c_1^+(s)(2\pi)^{-2} = 0$, i.e. $a_{-1} = \frac{i(k-2)}{2} a_{-2}$, which is the order- ε^{-1} term of $f(-1/\tau) = \tau^k f(\tau)$ at $\tau = i + \varepsilon$. $\square$

After projection, the surviving poles of $(I - \hat{P})f$ all lie strictly below the line $\text{im } \tau = 1$; since the heights of a Γ -orbit accumulate only at 0, when f has no pole with $\text{im } \tau_p = 1$ there is a maximal surviving height $y_b < 1$.

Lemma 4.10. *Let $f \in F_k$ have no pole with $\text{im } \tau_p = 1$; let $y_b < 1$ be the maximal height of a pole of $(I - \hat{P})f$, and m_b the maximal order among its poles at height y_b . Then there is a $C > 0$ such that, for every $\varepsilon > 0$,*

$$C n^{m_b-1} e^{2\pi n y_b} \leq |c_{Rf}(n)| \leq e^{2\pi n(y_b + \varepsilon)}, \quad (53)$$

the upper bound holding for all sufficiently large n , the lower for infinitely many n . In particular $|c_{Rf}(n)| e^{-2\pi n} \rightarrow 0$ geometrically, with base $e^{-2\pi(1-y_b-\varepsilon)}$ for every $\varepsilon > 0$.

Proof. $\square$

Classes are labelled by winding data: relative to a reference pair, let $X_S(p), X_T(p) \in \mathbb{Z}$ count the net turns of γ^S , resp. γ^T , about each pole $p \in \Gamma \cdot f^{-1}(\infty)$.

Lemma 4.11. *For $f \in F_k$,*

$$L^*(f, s; [\gamma_{\tau_0}^S], [\gamma_{\tau_0}^T]) = L_0^*(f, s) + 2\pi i e^{-\pi i s/2} \sum_p \left[X_S(p) \text{Res}_{\tau=p}(f(\tau) \tau^{s-1}) + X_T(p) \text{Res}_{\tau=p}(f(\tau) \tilde{k}_T(\tau, s)) \right], \quad (54)$$

with L_0^ the value at the reference pair: L_S^* shifts by the residues of $f \tau^{s-1}$, and L_T^* by those of $f \tilde{k}_T$.*

Proof. $\square$

Theorem 4.12. *Let $f \in F_k$ have no pole with $\text{im } \tau_p = 1$, and let $y_b < 1$ be as in Lemma 4.10. In the class of the pair at $\tau_0 = i$ — γ^T the straight segment $[i-1, i]$, γ^S contracted to the point i ($-1/i = i$); a convenient reference class, not a canonical one (Remark 4.13) —*

$$L^*(f, s) = -c_{Rf}(0) \left(\frac{1}{s} + \frac{i^k}{k-s} \right) + \sum_{n \neq 0} c_{Rf}(n) G_{s,k}(n) + \sum_{\text{im } \tau_p > 1} \sum_{m > 0} r_{f, \tau_p}^*(m) I_{m-1}(s, \tau_p), \quad (55)$$

and a general class adds the wall term of Lemma 4.11. Each sum converges exponentially: the n -sum with base $e^{-2\pi(1-y_b-\varepsilon)}$ (Lemma 4.10), each I_{m-1} with base $e^{-2\pi(\text{im } \tau_p - 1)}$ (Lemma 4.6). On the excluded line $\text{im } \tau_p = 1$ the tail integrals extend continuously away from the orbit of i (Lemma 4.7), and at i itself the wall value is that of Lemma 4.9; assembling these into the statement of the theorem at the wall is a principal-value bookkeeping we do not carry out here.

Proof. $\square$

Remark 4.13. *Coefficient by coefficient, $r_f|(1+S) = 0$ is the functional equation $L^*(f, s) = i^k L^*(f, k-s)$; the substantive further condition on a period polynomial is $r_f|(1+U+U^2) = 0$. For $f \in F_k$ both are properties of the homotopy class: by Lemma 4.11 the period polynomial*

shifts by residue polynomials under wall-crossing, and whether a given class satisfies either relation is arithmetic of those shifts. In a class satisfying both, r_f is a genuine Eichler–Shimura period polynomial, and such a class would anchor a canonical $L^*(f, s)$ for meromorphic f ; we take this question up elsewhere.

A Explicit period-polynomial bases

For each weight k with $\dim S_k = 1$, the even and odd rational period polynomials $W_{\pm}^{(k)}$ are the Kohnen–Zagier basis of the cuspidal period-polynomial space [6] — their leading coefficients are the Bernoulli numerators 691, 3617, 43867, $\dots$ — listed below with primitive integer coefficients. Alongside are the period polynomials of the cusp form $\Delta_k = f_{k,-1} = q + O(q^2)$ and the weak cusp form $\hat{\Delta}_k = f_{k,1} = q^{-1} + O(q^2)$, obtained from the finite-contour L -integral (Definition 2.3) and projected onto $W_{\pm}$ by the Haberland pairing $\langle P, Q \rangle = \sum_a (-1)^a \binom{n}{a}^{-1} p_a q_{n-a}$, $n = k - 2$. The periods $\omega^{\pm}(\Delta_k)$ and quasi-periods $\eta^{\pm}(\hat{\Delta}_k)$ are quoted to ten digits (full precision in [9], `period_polynomial_bases.py`); $r_{\hat{\Delta}_k}$ is the parabolic representative — the raw weak-form period polynomial differs by a C_T -coboundary the construction gauge-fixes.

Weight 12:

$$W_+^{(12)}(X, Y) = 36(X^{10} - Y^{10}) - 691(X^8 Y^2 - X^2 Y^8) + 2073(X^6 Y^4 - X^4 Y^6), \quad (56)$$

$$W_-^{(12)}(X, Y) = 4(X^9 Y + X Y^9) - 25(X^7 Y^3 + X^3 Y^7) + 42X^5 Y^5, \quad (57)$$

$$r_{\Delta_{12}} = 0.0001655268053 W_+^{(12)} + 0.009269276162 W_-^{(12)}, \quad (58)$$

$$r_{\hat{\Delta}_{12}} = -0.3061537142 W_+^{(12)} - 17.05597275 W_-^{(12)}. \quad (59)$$

Weight 16:

$$W_+^{(16)}(X, Y) = 360(X^{14} - Y^{14}) - 7234(X^{12} Y^2 - X^2 Y^{12}) + 25319(X^{10} Y^4 - X^4 Y^{10}) - 39787(X^8 Y^6 - X^6 Y^8), \quad (60)$$

$$W_-^{(16)}(X, Y) = 36(X^{13} Y + X Y^{13}) - 245(X^{11} Y^3 + X^3 Y^{11}) + 539(X^9 Y^5 + X^5 Y^9) - 660X^7 Y^7, \quad (61)$$

$$r_{\Delta_{16}} = 0.0002595173464 W_+^{(16)} + 0.01640585888 W_-^{(16)}, \quad (62)$$

$$r_{\hat{\Delta}_{16}} = -0.01227268476 W_+^{(16)} - 0.7278917244 W_-^{(16)}. \quad (63)$$

Weight 18:

$$W_+^{(18)}(X, Y) = 18000(X^{16} - Y^{16}) - 350936(X^{14} Y^2 - X^2 Y^{14}) + 1096675(X^{12} Y^4 - X^4 Y^{12}) - 1140542(X^{10} Y^6 - X^6 Y^{10}), \quad (64)$$

$$W_-^{(18)}(X, Y) = 24(X^{15} Y + X Y^{15}) - 154(X^{13} Y^3 + X^3 Y^{13}) + 273(X^{11} Y^5 + X^5 Y^{11}) - 143(X^9 Y^7 + X^7 Y^9), \quad (65)$$

$$r_{\Delta_{18}} = -3.122662695 \times 10^{-5} W_+^{(18)} - 0.1465501421 W_-^{(18)}, \quad (66)$$

$$r_{\hat{\Delta}_{18}} = -0.0001378466906 W_+^{(18)} - 0.907233517 W_-^{(18)}. \quad (67)$$

Weight 20:

$$\begin{aligned} W_+^{(20)}(X, Y) = & 26460(X^{18} - Y^{18}) - 523833(X^{16}Y^2 - X^2Y^{16}) \\ & + 1746110(X^{14}Y^4 - X^4Y^{14}) - 2444554(X^{12}Y^6 - X^6Y^{12}) \\ & + 2269943(X^{10}Y^8 - X^8Y^{10}), \end{aligned} \quad (68)$$

$$\begin{aligned} W_-^{(20)}(X, Y) = & 108(X^{17}Y + XY^{17}) - 715(X^{15}Y^3 + X^3Y^{15}) + 1452(X^{13}Y^5 + X^5Y^{13}) \\ & - 1560(X^{11}Y^7 + X^7Y^{11}) + 1430X^9Y^9, \end{aligned} \quad (69)$$

$$r_{\Delta_{20}} = 0.0001654739435 W_+^{(20)} + 0.2549699125 W_-^{(20)}, \quad (70)$$

$$r_{\hat{\Delta}_{20}} = -0.0002521273082 W_+^{(20)} + 0.1296973583 W_-^{(20)}. \quad (71)$$

Weight 22:

$$\begin{aligned} W_+^{(22)}(X, Y) = & 126000(X^{20} - Y^{20}) - 2485856(X^{18}Y^2 - X^2Y^{18}) \\ & + 8156715(X^{16}Y^4 - X^4Y^{16}) - 10564888(X^{14}Y^6 - X^6Y^{14}) \\ & + 6603055(X^{12}Y^8 - X^8Y^{12}), \end{aligned} \quad (72)$$

$$\begin{aligned} W_-^{(22)}(X, Y) = & 360(X^{19}Y + XY^{19}) - 2366(X^{17}Y^3 + X^3Y^{17}) + 4641(X^{15}Y^5 + X^5Y^{15}) \\ & - 4182(X^{13}Y^7 + X^7Y^{13}) + 1547(X^{11}Y^9 + X^9Y^{11}), \end{aligned} \quad (73)$$

$$r_{\Delta_{22}} = -0.0003341270695 W_+^{(22)} - 0.7346639014 W_-^{(22)}, \quad (74)$$

$$r_{\hat{\Delta}_{22}} = 0.0003197845612 W_+^{(22)} - 0.7633262888 W_-^{(22)}. \quad (75)$$

Weight 26:

$$\begin{aligned} W_+^{(26)}(X, Y) = & 34398000(X^{24} - Y^{24}) - 678984792(X^{22}Y^2 - X^2Y^{22}) \\ & + 2233675745(X^{20}Y^4 - X^4Y^{20}) - 2938319846(X^{18}Y^6 - X^6Y^{18}) \\ & + 2062613685(X^{16}Y^8 - X^8Y^{16}) - 850046852(X^{14}Y^{10} - X^{10}Y^{14}), \end{aligned} \quad (76)$$

$$\begin{aligned} W_-^{(26)}(X, Y) = & 1080(X^{23}Y + XY^{23}) - 7106(X^{21}Y^3 + X^3Y^{21}) + 14025(X^{19}Y^5 + X^5Y^{19}) \\ & - 13167(X^{17}Y^7 + X^7Y^{17}) + 7106(X^{15}Y^9 + X^9Y^{15}) - 1938(X^{13}Y^{11} + X^{11}Y^{13}), \end{aligned} \quad (77)$$

$$r_{\Delta_{26}} = -0.0002002972716 W_+^{(26)} - 40.08330039 W_-^{(26)}, \quad (79)$$

$$r_{\hat{\Delta}_{26}} = 0.0002002611462 W_+^{(26)} - 40.0894082 W_-^{(26)}. \quad (80)$$

References

- [1] K. Bringmann, K.-H. Fricke, and Z. Kent, *Special L-values and periods of weakly holomorphic modular forms*, Ramanujan J. **24** (2011), 119–139.
- [2] F. Brown, *A class of non-holomorphic modular forms III*, arXiv:1710.07912 (2017).
- [3] D. A. McGady, *L-functions for meromorphic modular forms and sum rules in conformal field theory*, arXiv:1806.09874 (2018).

- [4] T. M. Apostol, “Modular functions and Dirichlet series in number theory,” Graduate Texts in Mathematics, London, Springer (1997).
- [5] W. Duke and P. Jenkins, *On the zeros and coefficients of certain weakly holomorphic modular forms*, Pure Appl. Math. Q. **4** (2008), no. 4, 1327–1340.
- [6] W. Kohnen and D. Zagier, *Modular forms with rational periods*, in *Modular Forms* (R. A. Rankin, ed.), Ellis Horwood, 1984, pp. 197–249.
- [7] F. Brown and R. Hain, *Algebraic de Rham theory for weakly holomorphic modular forms of level one*, Algebra Number Theory **12** (2018), no. 3, 723–750; arXiv:1708.00652.
- [8] H. Cohen, *Haberland’s formula and numerical computation of Petersson scalar products*, in *ANTS X: Proceedings of the Tenth Algorithmic Number Theory Symposium*, Open Book Ser. **1**, Math. Sci. Publ., 2013, pp. 249–270.
- [9] *Numerical verification scripts* (companion code), 2026. Published checks: `elliptic_period_checks.py`; full working ledger: `wip_meromorphic_periods.py`.
- [10] NIST Digital Library of Mathematical Functions, F. W. J. Olver *et al.*, eds., <https://dlmf.nist.gov/>, Release 1.2.x. Chapters used: §8 (incomplete gamma) and §25.11 (Hurwitz zeta).
- [11] R. Bruggeman, Y. Choie, and N. Diamantis, *Holomorphic automorphic forms and cohomology*, Mem. Amer. Math. Soc. **253** (2018), no. 1212; arXiv:1404.6718.
- [12] K. Bringmann, P. Guerzhoy, Z. Kent and K. Ono, *Eichler-Shimura Theory for Mock Modular Forms*, Mathematische Annalen, March 2013 355(3).
- [13] V. Paşol and A. A. Popa, *Modular forms and period polynomials*, Proc. London Math. Soc. (3) **107** (2013), 713–743; arXiv:1202.5802.
- [14] W. Wang and H. Zhang, *Meromorphic quasi-modular forms and their L-functions*, arXiv:2202.09542 (2022).
- [15] F. Brown and T. J. Fonseca, *Single-valued periods of meromorphic modular forms and a motivic interpretation of the Gross–Zagier conjecture*, arXiv:2508.04844 (2025).
- [16] K. Bringmann, N. Diamantis, and S. Ehlen, *Regularized inner products and errors of modularity*, Int. Math. Res. Not. IMRN (2016); arXiv:1603.03056.
- [17] N. Diamantis and J. Drewitt, *Period functions associated to real-analytic modular forms*, Res. Math. Sci. **7** (2020), no. 21.